

# Prakhar Batwal

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## Education

Parul University — B.Tech, Computer Science & Engineering

Aug 2023 – Apr 2027 | Vadodara, India

## Technical Skills

- **ML & Deep Learning:** TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, FAISS, SentenceTransformers, SHAP, NLTK
- **MLOps & DevOps:** Docker, MLflow, DagsHub, GitHub Actions, Git, FastAPI, MongoDB, Redis, MySQL, PostgreSQL
- **Languages:** Python, Java, SQL (MySQL, PostgreSQL), JavaScript (ES6+)
- **CS Fundamentals:** Data Structures & Algorithms, System Design, OOP, DBMS, Operating Systems, Computer Networks

## Projects

### Goodreads Recommendation System — Neural Search & Two-Tower Engine

Apr 2026 – Present

Python · TensorFlow/Keras · FAISS · SentenceTransformers · FastAPI · Docker

GitHub

- Achieved **Hit Rate@10 of 25.2%** and **MRR 0.1078** by training a two-tower collaborative filtering network on 200K+ ratings, projecting users and items into a shared 64-D latent space.
- Replaced  $O(N)$  brute-force retrieval with a **FAISS IndexFlatIP** index backed by **SentenceTransformers** real-time encoding, cutting retrieval complexity to  $O(\log N)$  ANN lookup.
- Refactored the vectorization pipeline to embed unique catalog records once rather than per-transaction rows, cutting **CPU preprocessing time by 95%** and eliminating memory pressure at scale.
- Deployed async **FastAPI** serving with **Pydantic** validation, containerized via **Docker**, and automated integration tests on every merge via **GitHub Actions** CI.

### Real Estate Valuation Engine — Ensemble Modeling & SHAP Interpreter

Dec 2025 – Jan 2026

Python · XGBoost · LightGBM · SHAP · scikit-learn · FastAPI · SQLite · Docker

GitHub

- Reached  **$R^2 = 0.847$**  on California housing data with a **VotingRegressor** (XGBoost + LightGBM) augmented by custom geospatial features and **KMeans** neighborhood clustering.
- Delivered **sub-50 ms inference latency** through a Dockerized async **FastAPI** service with strict **Pydantic** validation handling concurrent prediction load reliably.
- Appended per-feature **SHAP USD contributions** from a TreeExplainer pipeline to every API response, making ensemble decisions fully auditable without additional tooling overhead.
- Persisted predictions and top SHAP drivers to **SQLite** for drift monitoring; automated regression tests on every push via a **GitHub Actions** CI/CD pipeline.

### TruthLens — AI News Credibility & Bias Analyzer

Aug 2025

Python · scikit-learn · TF-IDF · NLTK · OpenAI API · FastAPI · pytest

GitHub

- Trained a **dual-output classifier** (binary credibility + 5-class political bias) with **TF-IDF** and grid-search sweeps in scikit-learn, optimizing F1 on imbalanced news corpora.
- Layered **NLTK VADER** for sentence-level sentiment scoring and wired async **OpenAI API** calls for token-level linguistic bias breakdowns alongside classifier confidence scores.
- Surfaced predictions through a **JavaScript/HTML5** dashboard with bias distribution charts, confidence heatmaps, and per-token sentiment highlights for non-technical users.
- Enforced **zero regressions** at launch via full **pytest** suite across all modules (*services, utils, models*) with strict modular package layout separating concerns cleanly.

### Network Security — Phishing Detection & End-to-End ML Pipeline

Jun – Jul 2025

Python · scikit-learn · MongoDB · MLflow · DagsHub · FastAPI · PyYAML

GitHub

- Architected a **4-stage modular pipeline** (Ingestion → Validation → Transformation → Training) for phishing URL classification with fully automated **MongoDB** data seeding.
- Built a **PyYAML schema validation** layer detecting feature anomalies and data drift pre-training, applying **KNN Imputation** and robust scaling on noisy real-world network data.
- Tracked **Random Forest, Gradient Boosting, and AdaBoost** across hyperparameter sweeps via **MLflow + DagsHub**, enabling reproducible comparison and one-click model rollback.
- Deployed a batch **FastAPI + Uvicorn** inference API accepting raw CSV payloads and returning dynamically rendered **HTML prediction reports**, reducing analyst triage overhead.

## Achievements & Profiles

- **Elevate Hackathon — Top 10 / 700+ teams (top 1.4%):** Shipped a production-ready ML application in 48 hours under competitive constraints.
- **Competitive Programming:** 400+ problems on LeetCode & CodeChef; medium-hard difficulty, sub-500 ms Java solutions.
- **Engineering Leadership:** Led 4-person team across 2 projects — Git strategy, PR reviews, zero post-launch critical bugs.
- **YouTube Tech Channel:** Publishes full-stack & ML deployment tutorials demonstrating technical communication depth beyond code.